Name: Pakin Panawattanakul ID: 6580043

Exercise 7 (14 points) – individual work

- The answers can be typed or handwritten (handwriting must be clear and readable), in this exercise sheet or your own sheet (put your name & ID at the top of the sheet). All answers must be saved to only 1 PDF file.
- Some questions also require the submission of processes/workflows (file.rmp or file.ipynb).
- In case of re-submission (after first grading) or submission after solution is given, your points will be weighted by 0.5.

Retrieve **breakfast** data for association analysis. The data is in transaction format. Each record consists of guest ID + guest's selections from a hotel's breakfast buffet. There are 8 dishes being offered: cold cut, dimsum, muesli yogurt, omelette, pasta salad, padthai, porridge, sandwich. Generate frequent itemsets and association rules.

1. (3 points) Find the following itemsets and their supports:

(a) 1-itemset containing porridge	support = 0.533
(b) Among all 2-itemsets containing porridge, pick the one(s) with max support	2-itemset = (porridge, dimsum) support = 0.333
(c) Among all 3-itemsets containing porridge, pick the one(s) with max support	3-itemset = (omelette, porridge, dimsum) support = 0.200

2. (4 points) Consider the following rules. Report/capture each rule with support, confidence, and lift metrics. Interpret each metric and draw conclusion about the guests' breakfast preferences.

<table>
<tr>
<td>(a) Rule "omelette ⍰ muesli yogurt"</td>
<td>
Show rule with metrics
<table>
<tr><th>Premises</th><th>Conclusion</th><th>Support</th><th>Confidence</th><th>Lift</th></tr>
<tr><td>omelette</td><td>muesli yogurt</td><td>0.200</td><td>0.353</td><td>0.814</td></tr>
</table>
Interpretation
support(A->B) = P(A,B)
Support = 0.2 →20 % of guests eat omelette and muesli yogurt together

Confidence(A->B) : support(A,B)/support(A) = P(A,B)/P(A) = P(B|A)
Confidence = 0.353 → if guests eat omelette there are 35.3% chance that he will eat muesli yogurt

lift(A->B) : Confidence(A->B)/ support(B)
Lift = 0.814 < 1 negative association if guests eat omelette there are less chance that he/she will also eat muesli yogurt and viseversa
</td>
</tr>
</table>

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<table>
<tr>
<td>(b) Rule “dimsum porridge”</td>
<td>Show rule with metrics
<table>
<tr><th>Premises</th><th>Conclusion</th><th>Support</th><th>Confide... ↓</th><th>Lift</th></tr>
<tr><td>dimsum</td><td>porridge</td><td>0.333</td><td>0.667</td><td>1.250</td></tr>
</table>
Interpretation
support(A->B) = P(A,B)
Support = 0.333 →33.33% of guests eat dimsum and porridge together

Confidence(A->B) : support(A,B)/support(A) = P(A,B)/P(A) = P(B|A)
Confidence = 0.667 → if a guess eat dimsum there are 66.7% chance that he will eat porridge

lift(A->B) : Confidence(A->B)/ support(B)
Lift = 1.250 > 1 positive association if guess eat dimsum there is more chance that he/she will also eat porridge and viseversa</td>
</tr>
</table>

3. (2 points) Compare the following rules. Report/capture each rule with support, confidence, and lift metrics. And based on these metrics, which rule has higher quality?

<table>
<tr>
<td>(a) Rule “muesli yogurt, pasta salad sandwich”</td>
<td>Show rule with metrics
<table>
<tr><td>muesli yogurt, pasta salad</td><td>sandwich</td><td>0.100</td><td>0.500</td><td>1.875</td></tr>
</table>
Support = 0.100
Confidence = 0.500
Lift = 1.875</td>
</tr>
<tr>
<td>(b) Rule “sandwich muesli yogurt, pasta salad”</td>
<td>Show rule with metrics
<table>
<tr><td>omelette</td><td>muesli yogurt, pasta salad</td><td>0.100</td><td>0.176</td><td>0.882</td></tr>
</table>
Support = 0.100
Confidence = 0.176
Lift = 0.862</td>
</tr>
<tr>
<td>(c) Which rule has higher quality?</td>
<td>Rule “muesli yogurt, pasta salad sandwich” since they have gher confidence and lift</td>
</tr>
</table>

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4. (3 point) Suppose that we want to relocate padthai counter. Choose 1 other food counter that padthai should stay closest to. Your answer must be supported by at least 1 association rule. Report/capture that rule with its support, confidence, lift. Also give a brief explanation to justify your choice.

Premises	Conclusion	Support ↓	Confidence	Lift
muesli yogurt	padthai	0.100	0.231	1.731

Muesli yogurt would be the choice since it is the highest support from all the rules that have a conclusion as pad thai. The confidence is a bit low but it is better when compared with slightly higher confidence but much lower support. And high lift show positive association

5. (2 points) Suppose that we want to convert this dataset to fixed-length format. How many new attributes (excluding guest ID) do we need for the new format? And should we treat them as symmetric or asymmetric attributes?

Asymmetric attributes with 8 new attributes one for each for a dish name. And the value would be {0,1} for presence and absence of the dish. The presence is informative and absence is not.c